

Composition of two Simple Harmonic Vibrations



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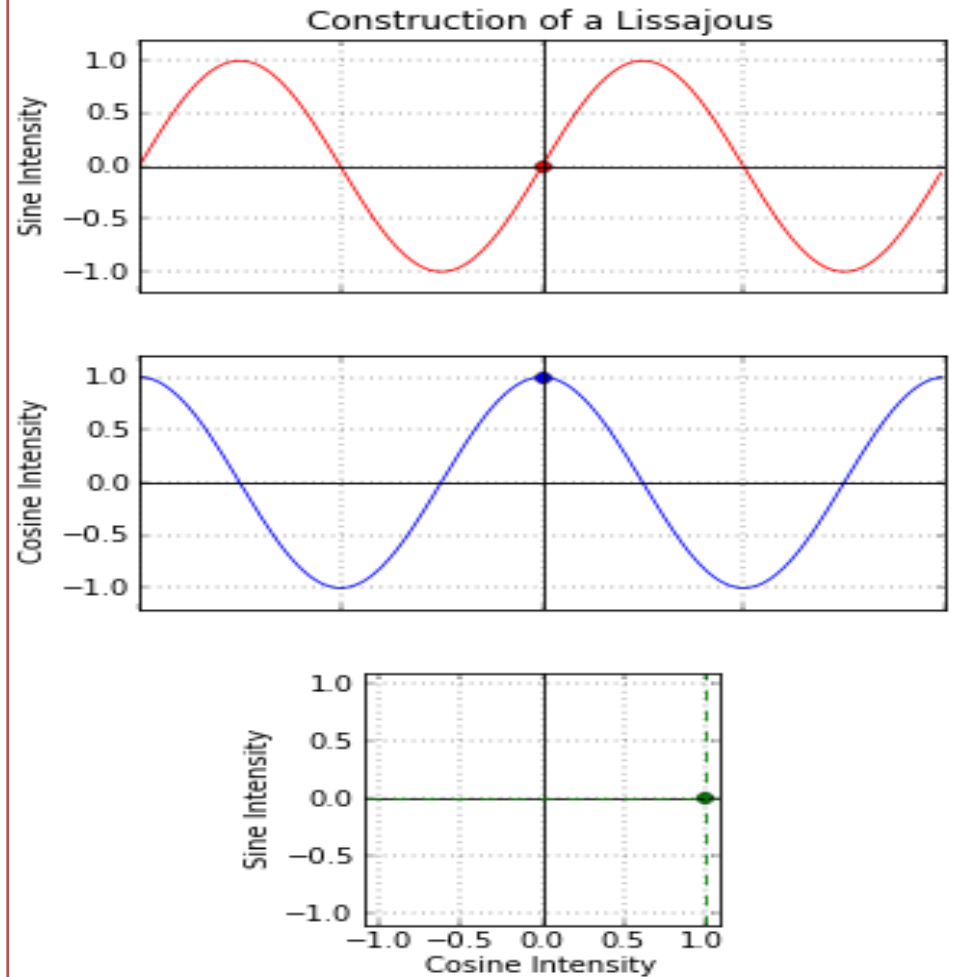
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Whatever we will discuss

- ✓ Lissajous Figures
- ✓ Composition of two Simple Harmonic Vibrations
- ✓ Related mathematical problems

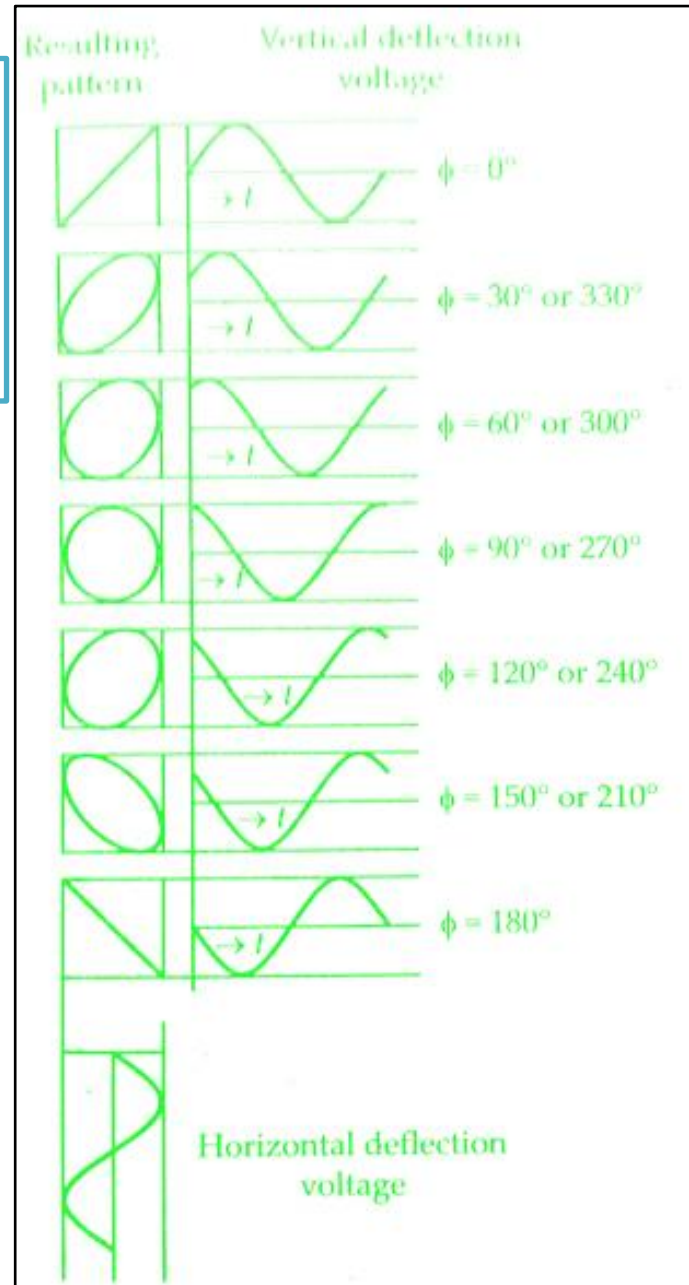
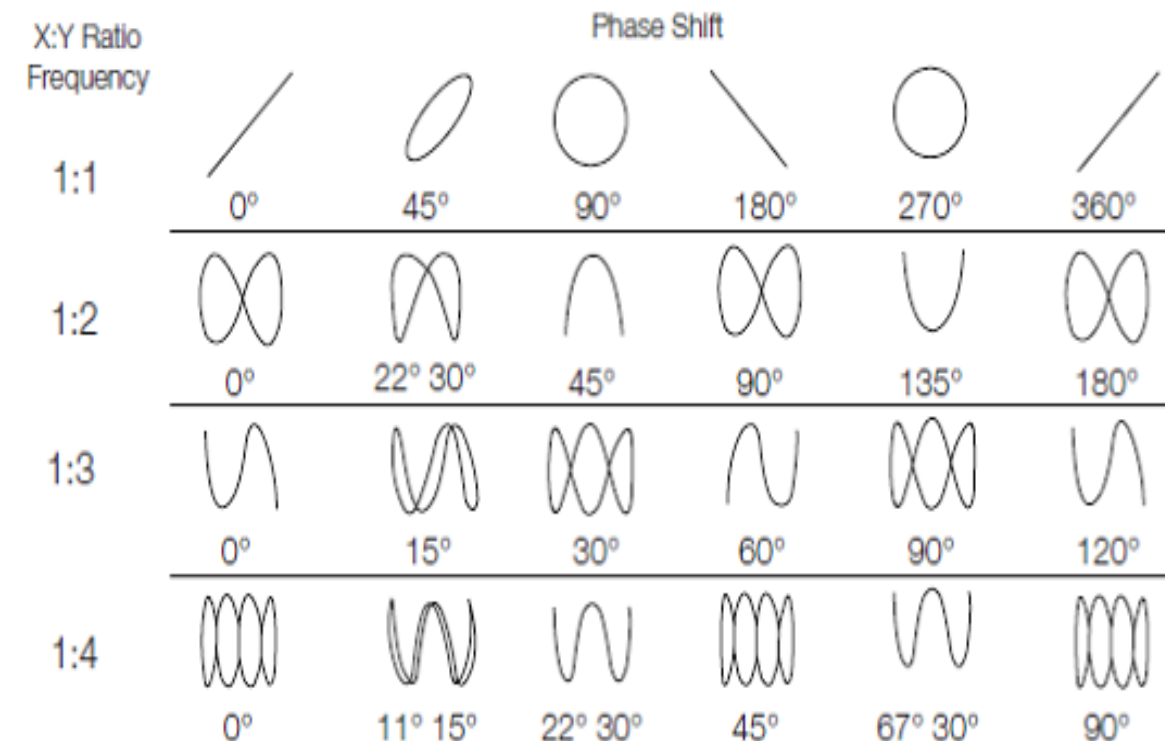
Lissajous Figures

➤ The simultaneously influenced of two simple harmonic motions at right angle to each other, the resultant motion of the particle traces a curve, which is known as Lissajous Figures.



Lissajous Figures

If two S.H.M's act in perpendicular directions, then their resultant motion is in the form of a straight line or a circle or a parabola etc. depending on the frequency ratio of the two S.H.M. and initial phase difference. These figures are called Lissajous figures.



Composition of two SHM's at right angles to each other having same frequencies but differing in phases and amplitudes

Let us consider two simple harmonic vibrations, which are represented by the equations

$$x = a \sin(\omega t + \phi) \quad \dots \quad \dots \quad (1) \qquad \text{From Equ. (2)}$$

$$y = b \sin \omega t \quad \dots \quad \dots \quad \dots \quad (2) \qquad \sin \omega t = \frac{y}{b}$$

From Equ. (1)

$$\begin{aligned} \frac{x}{a} &= \sin(\omega t + \phi) \\ &= \sin \omega t \cos \phi + \cos \omega t \sin \phi \\ &= \sin \omega t \cos \phi + \sqrt{(1 - \sin^2 \omega t)} \sin \phi \\ &= \frac{y}{b} \cos \phi + \sqrt{(1 - \frac{y^2}{b^2})} \sin \phi \end{aligned}$$

$$\frac{x}{a} - \frac{y}{b} \cos \phi = \sqrt{(1 - \frac{y^2}{b^2})} \sin \phi \quad \dots \quad \dots \quad (3)$$

Squaring both sides of Equ. (3)

$$\begin{aligned} \frac{x^2}{a^2} + \frac{y^2}{b^2} \cos^2 \phi - 2 \frac{x}{a} \frac{y}{b} \cos \phi &= (1 - \frac{y^2}{b^2}) \sin^2 \phi \\ \frac{x^2}{a^2} + \frac{y^2}{b^2} \cos^2 \phi + \frac{y^2}{b^2} \sin^2 \phi - 2 \frac{x}{a} \frac{y}{b} \cos \phi &= \sin^2 \phi \\ \frac{x^2}{a^2} + \frac{y^2}{b^2} - 2 \frac{xy}{ab} \cos \phi &= \sin^2 \phi \dots \dots \dots (4) \end{aligned}$$

Equ. (4) gives the general equations of the resultant vibration of the two vibrations given by Equ. (1) and Equ. (2).

Composition of two SHM's at right angles to each other having same frequencies but differing in phases and amplitudes

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - 2 \frac{xy}{ab} \cos \phi = \sin^2 \phi \dots \dots (4)$$

Equ. (4) is the general equations of a conic whose shape will depend upon the value of the phase difference ϕ between two vibrations.

Case-I: $\phi = 0, 2\pi, 4\pi, \dots = 2n\pi$;

Equ. (4) becomes, $y = \frac{b}{a} x$; which is the equation of a straight line.

Case-II: $\phi = \pi/4$ radian or 45° ;

Equ. (4) becomes, $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{\sqrt{2}xy}{ab} = \frac{1}{2}$;

which represents the equation of an oblique ellipse.

Case-III: $\phi = \pi/2$ radian or 90° ;

Equ. (4) becomes, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$;

which represents the equation of a symmetrical ellipse.

Composition of two SHM's at right angles to each other having same frequencies but differing in phases and amplitudes

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - 2 \frac{xy}{ab} \cos \phi = \sin^2 \phi \dots \dots (4)$$

Equ. (4) is the general equations of a conic whose shape will depend upon the value of the phase difference ϕ between two vibrations.

Case-IV: $\phi = 3\pi/4$ radian or 135° ;

Equ. (4) becomes, $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{\sqrt{2}xy}{ab} = \frac{1}{2}$;

which represents the equation of an oblique ellipse.

Case-V: $\phi = \pi$ radian or 180° ;

Equ. (4) becomes, $y = -\frac{b}{a}x$; which is the equation of a straight line.

Composition of two Collinear Simple harmonic vibrations of same frequency but differing in phases and amplitudes

Let us two simple harmonic vibrations be represented by the equations

$$y_1 = a_1 \sin(\omega t + \alpha_1) \quad \dots \quad \dots \quad (i)$$

$$y_2 = a_2 \sin(\omega t + \alpha_2) \quad \dots \quad \dots \quad (ii)$$

The resultant displacement y of the particle is given by

$$y = y_1 + y_2 = a_1 \sin(\omega t + \alpha_1) + a_2 \sin(\omega t + \alpha_2)$$

Composition of two simple harmonic vibrations

$$= a_1(\sin \omega t \cos \alpha_1 + \cos \omega t \sin \alpha_1) + a_2(\sin \omega t \cos \alpha_2 + \cos \omega t \sin \alpha_2)$$

$$y = (a_1 \cos \alpha_1 + a_2 \cos \alpha_2) \sin \omega t + (a_1 \sin \alpha_1 + a_2 \sin \alpha_2) \cos \omega t \quad \dots (iii)$$

The amplitudes and angles are constant, so

$$A \cos \phi = a_1 \cos \alpha_1 + a_2 \cos \alpha_2 \quad \dots \dots (iv)$$

$$A \sin \phi = a_1 \sin \alpha_1 + a_2 \sin \alpha_2 \quad \dots \dots (v)$$

Now squaring and adding equations (iv) and (v), we get

Composition of two simple harmonic vibrations

$$A^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos(\alpha_1 - \alpha_2) \quad \dots \quad \text{..(vi)}$$

$$\tan \phi = \frac{a_1 \sin \alpha_1 + a_2 \sin \alpha_2}{a_1 \cos \alpha_1 + a_2 \cos \alpha_2} \quad \dots \quad \dots \quad \text{(vii)}$$

The resultant displacement

$$y = A \cos \phi \sin \omega t + A \sin \phi \cos \omega t = A \sin(\omega t + \phi) \quad \dots \quad \dots \quad \dots \quad \text{(viii)}$$

Equation (vi) and (vii) gives the value of A and ϕ . Equation (viii) is similar to the original equation (i) and (ii).

Composition of two Collinear Simple harmonic vibrations

Special Cases

Same Phase: If the two simple harmonic vibrations are in the same phase, then $\alpha_1 = \alpha_2 = \alpha$ (say).

The phase difference

$$(\alpha_1 - \alpha_2) = 0, 2\pi, 4\pi, \dots = 2n\pi. \text{ where } n = 0, 1, 2, \dots$$

$$\cos(\alpha_1 - \alpha_2) = 1$$

$$A^2 = a_1^2 + a_2^2 + 2a_1a_2 = (a_1 + a_2)^2 \quad \text{So, } A = a_1 + a_2$$

$$\tan \phi = \frac{(a_1 + a_2) \sin \alpha}{(a_1 + a_2) \cos \alpha} = \tan \alpha$$

Composition of two simple harmonic vibrations

Opposite phase: If the two simple harmonic vibrations acting on the particle are in the opposite phase

The phase difference

$$(\alpha_1 - \alpha_2) = \pi, 3\pi, 5\pi, \dots = (2n + 1)\pi. \text{ where } n = 0, 1, 2, \dots$$

$$\cos(\alpha_1 - \alpha_2) = -1$$

$$A^2 = a_1^2 + a_2^2 - 2a_1a_2 = (a_1 - a_2)^2$$

$$\text{So, } A = a_1 - a_2$$

Problem _ Solution

Two simple harmonic motions acting simultaneously on a particle are given by the equations

$$y_1 = 2 \sin (\omega t + \pi/6)$$

$$y_2 = 3 \sin (\omega t + \pi/3)$$

Calculate (i) amplitude, (ii) phase constant and (iii) time period of the resultant vibration. What is the equation of the resultant vibration?

Problem _ Solution

Solution:

$$y_1 = 2 \sin (wt + \pi/6) \dots\dots\dots (i)$$

$$y_2 = 3 \sin (wt + \pi/3) \dots\dots\dots (ii)$$

The equations are similar to the equation

$$y_1 = a_1 \sin (wt + \alpha_1) \dots\dots\dots (iii)$$

$$y_2 = a_2 \sin (wt + \alpha_2) \dots\dots\dots (iv)$$

The equation of the resultant vibration is given by

$$y = A \sin (wt + \phi) \dots\dots\dots (v)$$

$$\text{where } A^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos (\alpha_1 - \alpha_2) \dots\dots\dots (vi)$$

$$\text{and } \phi = \tan^{-1} \left(\frac{a_1 \sin \alpha_1 + a_2 \sin \alpha_2}{a_1 \cos \alpha_1 + a_2 \cos \alpha_2} \right) \dots\dots\dots (vii)$$

$$\text{here } a_1 = 2, a_2 = 3, \alpha_1 = \pi/6, \alpha_2 = \pi/3$$

Problem _ Solution

(i) The resultant amplitude

$$\begin{aligned} A &= \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos(\alpha_1 - \alpha_2)} \\ &= \sqrt{4 + 9 + (2)(2)(3)\cos\left(-\frac{\pi}{6}\right)} \\ &= 4.939 \end{aligned}$$

$$\begin{aligned} \text{(ii) } \phi &= \tan^{-1} \left[\frac{(2)(0.5) + (3)(0.866)}{(2)(0.866) + (3)(0.5)} \right] \\ &= \tan^{-1}[1.114] \\ &= 48.1^\circ \simeq \frac{4\pi}{15} \end{aligned}$$

$$\therefore \text{phase constant} = \left(\omega t + \frac{4\pi}{15} \right)$$

(iii) The resultant time period is the same as the time period of the individual vibrations. The equation of the resultant vibration is

$$y = 4.939 \sin \left(\omega t + \frac{4\pi}{15} \right)$$

Uses of lissajous figures

To determine the ratio of time period

To determine the frequency of a tuning fork



Thank You
Thanks for your attention